

ELEMENT COMBINATION **RULE**

CIRCUIT ANALYSIS METHOD

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TOPIC OUTLINE

Ohm's Law

Series Network

Parallel Network

Series-Parallel Network



OHM'S LAW



OHM'S LAW

Ohm's Law states that the ratio of voltage (V) to current (I) is **constant** (R).

Mathematical Representation

$$R = \frac{V}{I}$$

Basic Electrical Quantities

1. **Voltage (V)**

The measure of electrical potential energy per unit charge. It is the "**push**" or "force" that drives electric current through a circuit.

Formula

$$V = IR$$

unit

Volt (V)



OHM'S LAW

Ohm's Law states that the ratio of voltage (V) to current (I) is **constant** (R).

Mathematical Representation

$$R = \frac{V}{I}$$

Basic Electrical Quantities

2. **Current (I):**

The flow of electric charge, typically carried by electrons in a conductor. It represents the rate at which charge flows through a point in a circuit.

Formula

$$I = \frac{V}{R}$$

unit

Ampere (A)



OHM'S LAW

Ohm's Law states that the ratio of voltage (V) to current (I) is **constant** (R).

Mathematical Representation

$$R = \frac{V}{I}$$

Basic Electrical Quantities

3. **Resistance (R)**

The **opposition** to the flow of electric current in a material or component. It determines how much current will flow for a given voltage.

Formula

$$R = \frac{V}{I}$$

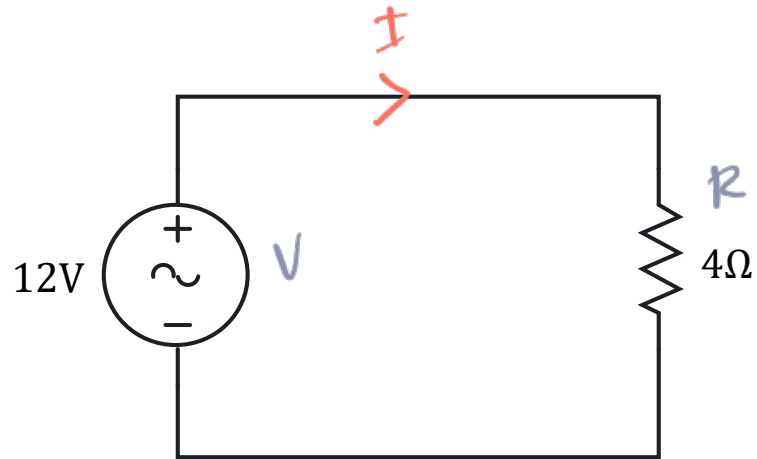
unit

Ohm (Ω)



EXERCISE

Determine the current flowing through a circuit with 4 ohms resistance when a voltage of 12 volts is applied.



Solution

$$I = \frac{V}{R}$$

$$I = \frac{12}{4}$$

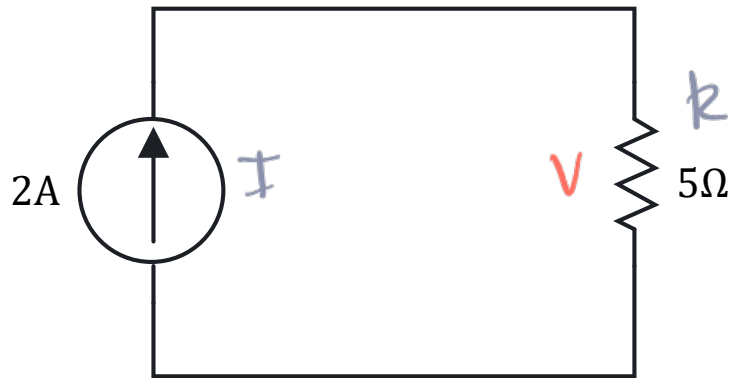
$$I = 3A$$

ans



EXERCISE

Determine the voltage drop across a circuit with a resistance of 5 ohms when a current of 2 amps flows through it.



Solution

$$V = IR$$

$$V = 2(5)$$

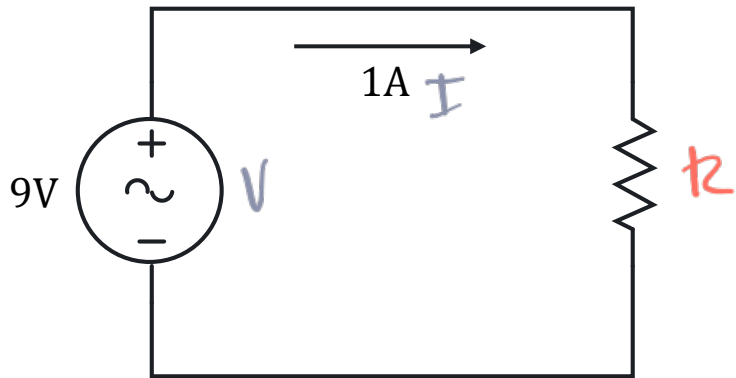
$$V = 10V$$

ans



EXERCISE

Determine the resistance of a circuit if a 9-volt applied voltage results in a current flow of 1-ampere.



Solution

$$R = \frac{V}{I}$$

$$R = \frac{9}{1}$$

$$R = 9 \Omega$$

ans

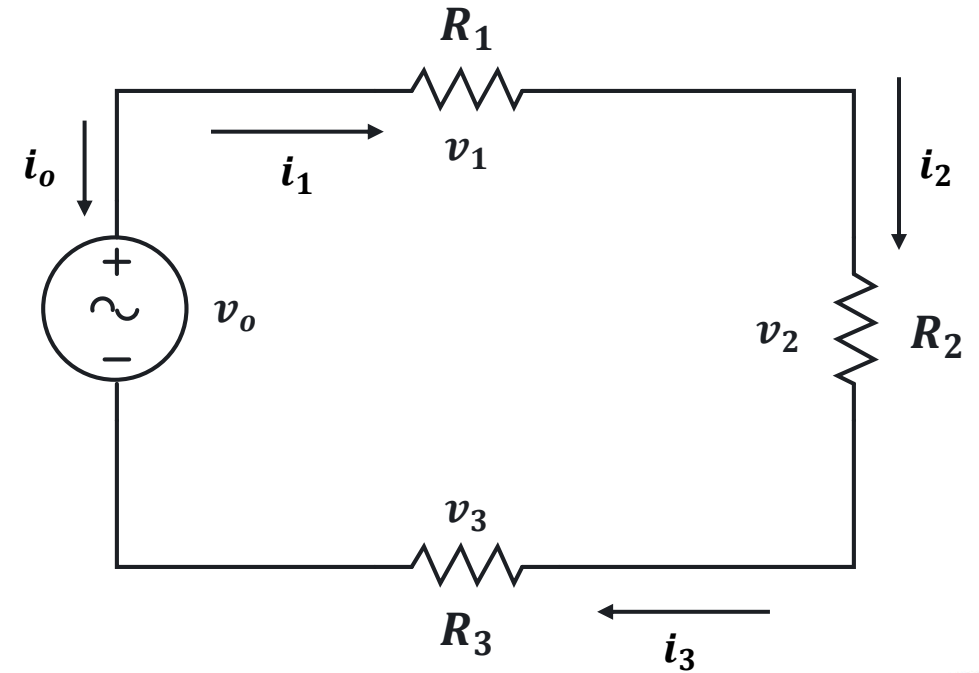
SERIES NETWORK



SERIES NETWORK

A series network refers to a configuration where components are connected end-to-end, forming a single path for current to flow.

Series Network



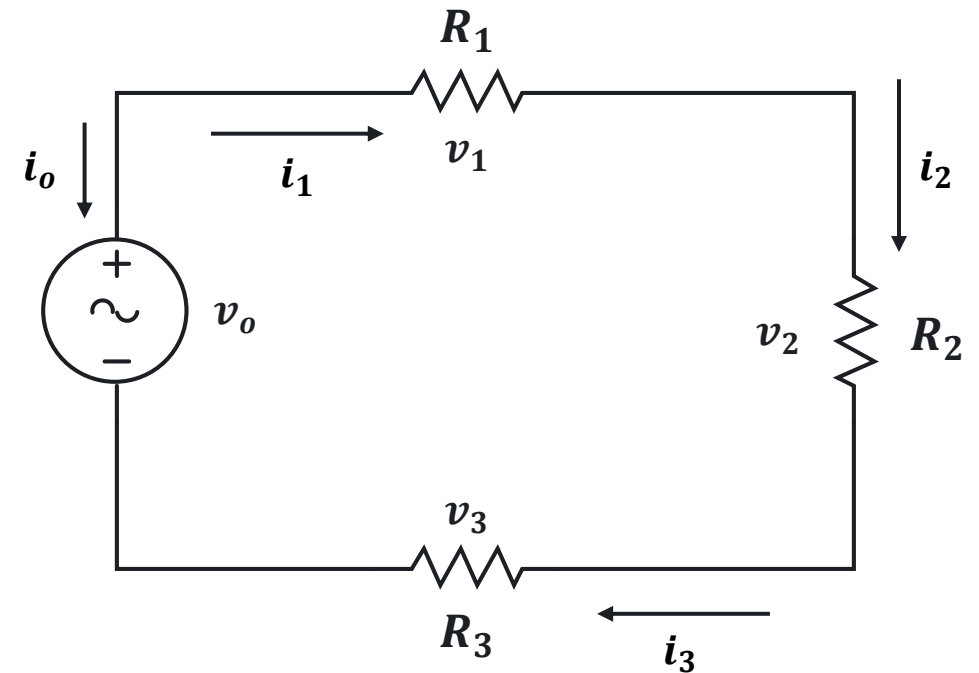
CURRENT

In a series network, the same current flows through all components.

Mathematical Representation

$$i_o = i_1 = i_2 = i_3 = \cdots i_n$$

Series Network



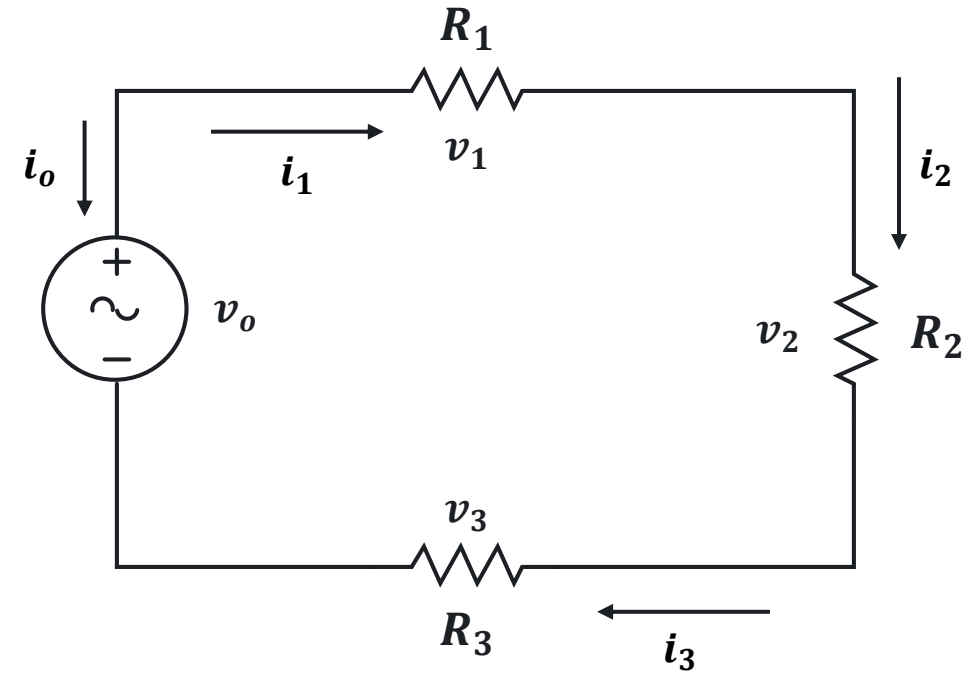
RESISTANCE

In a series network, the total resistance is the sum of the individual resistances.

Mathematical Representation

$$R_o = R_1 + R_2 + R_3 + \cdots R_n$$

Series Network



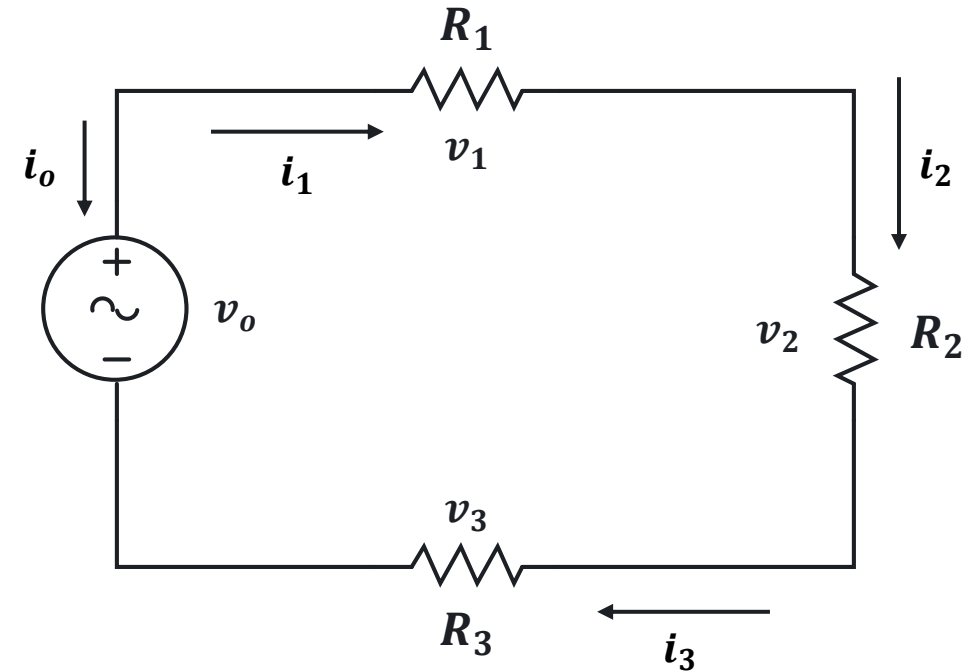
VOLTAGE

In a series network, the total voltage is the sum of the voltages across each individual component.

Mathematical Representation

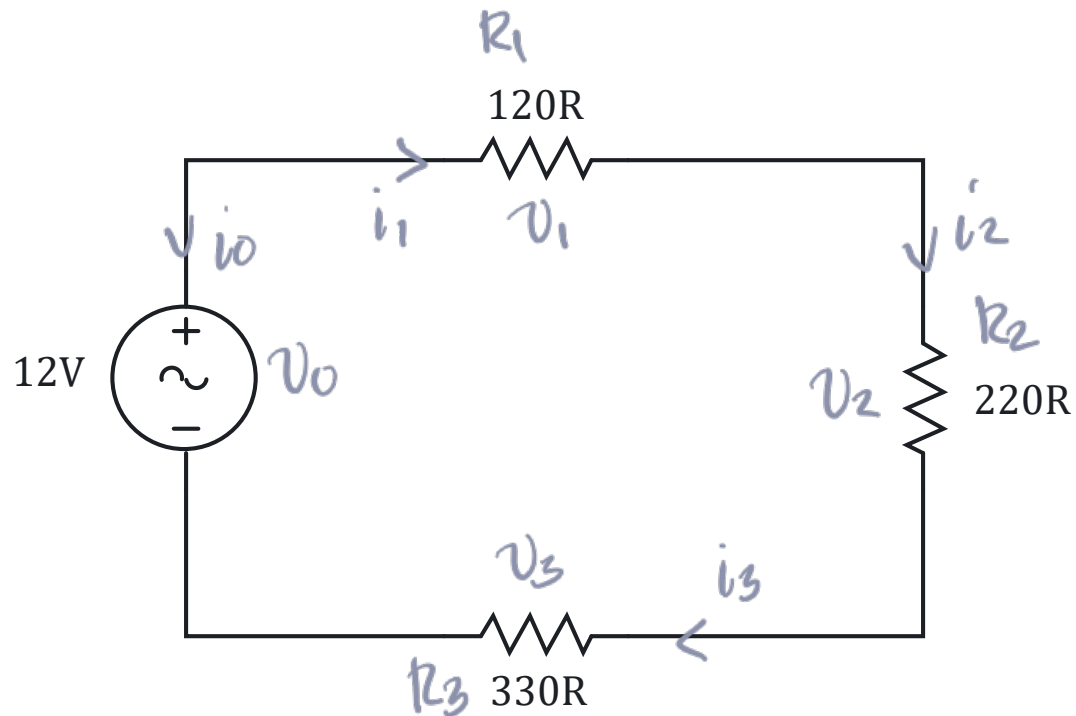
$$v_o = v_1 + v_2 + v_3 + \cdots v_n$$

Series Network



EXERCISE

Determine the **voltage drop** across each resistor of the given circuit.



Solution

Total Resistance

$$R_0 = R_1 + R_2 + R_3$$

$$R_0 = 120 + 220 + 330$$

$$R_0 = 670\Omega$$

Total Current

$$i_0 = \frac{V_0}{R_0}$$

$$i_0 = i_1 = i_2 = i_3$$

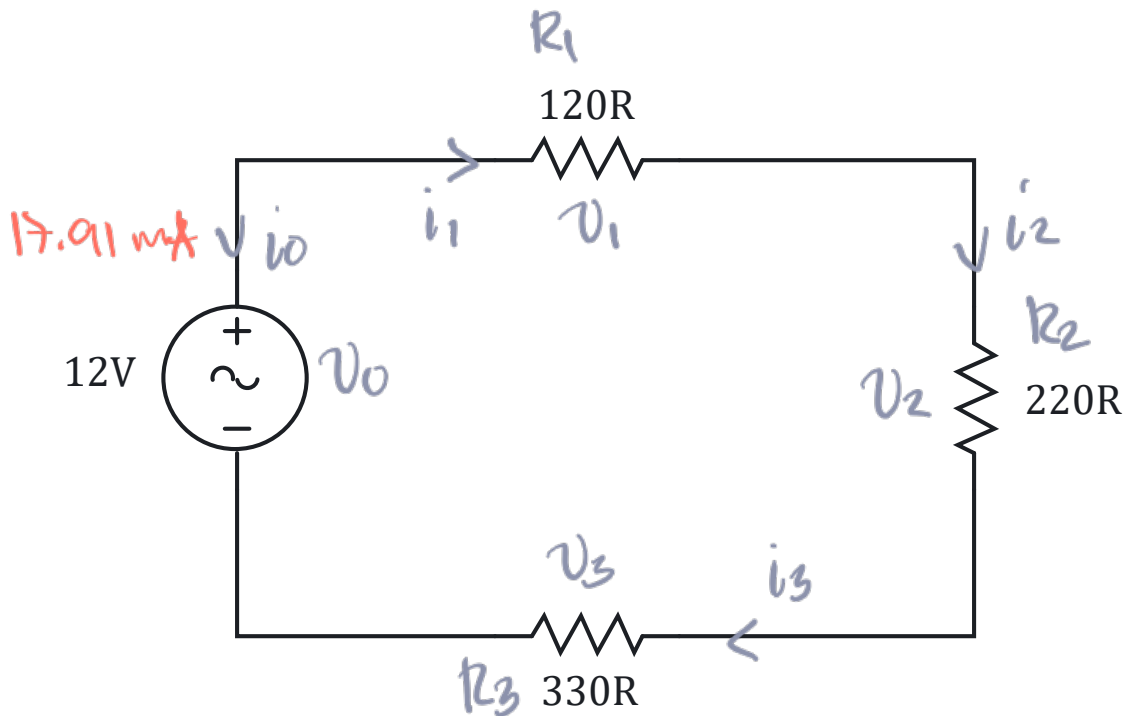
$$i_0 = \frac{12}{670}$$

$$i_0 = 17.91\text{ mA}$$



EXERCISE

Determine the **voltage drop** across each resistor of the given circuit.



Solution

$$V_1 = i_1 R_1$$

$$V_1 = 17.91 \text{ m}(120)$$

$$V_1 = 2.15 \text{ V}$$

ans

$$V_3 = i_3 R_3$$

$$V_3 = 17.91 \text{ m}(330)$$

$$V_3 = 5.91 \text{ V}$$

ans

$$V_2 = i_2 R_2$$

$$V_2 = 17.91 \text{ m}(220)$$

$$V_2 = 3.94 \text{ V}$$

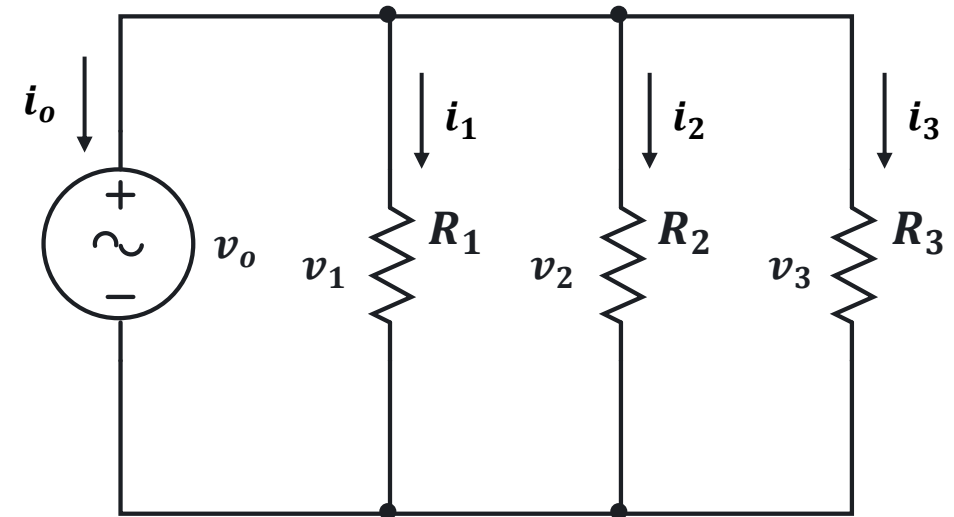
ans



PARALLEL NETWORK

A parallel network is a configuration where components are connected across the same two points, providing multiple paths for current to flow.

Parallel Network



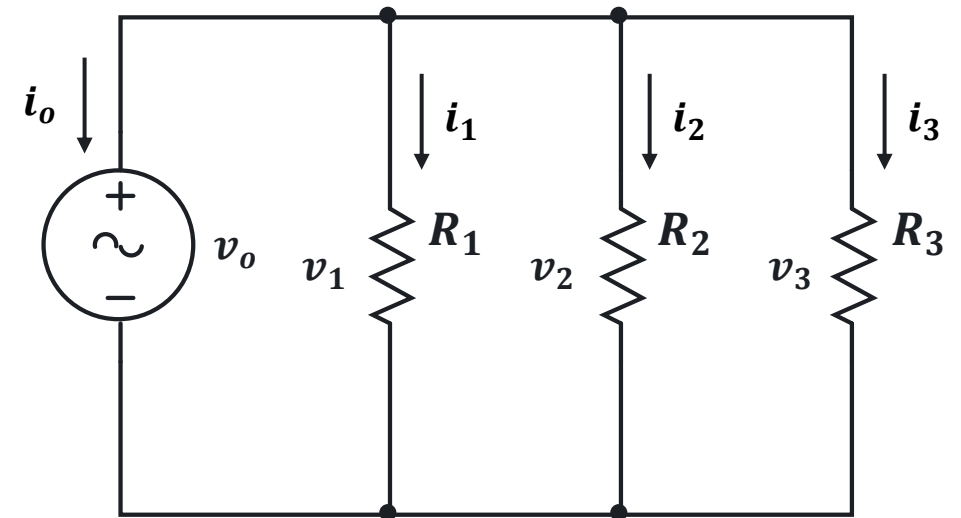
VOLTAGE

In a parallel network, the voltage across all components is the same.

Mathematical Representation

$$v_o = v_1 = v_2 = v_3 = \cdots v_n$$

Parallel Network



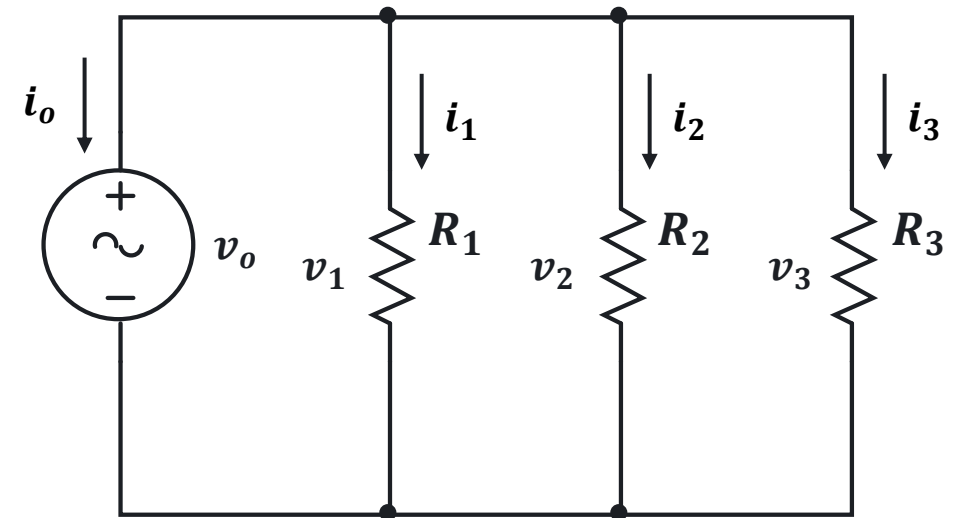
CONDUCTANCE

Conductance refers to the ability of the network to allow the flow of electric current. It is the reciprocal of resistance and is measured in siemens (S).

Mathematical Representation

$$G = \frac{1}{R}$$

Parallel Network



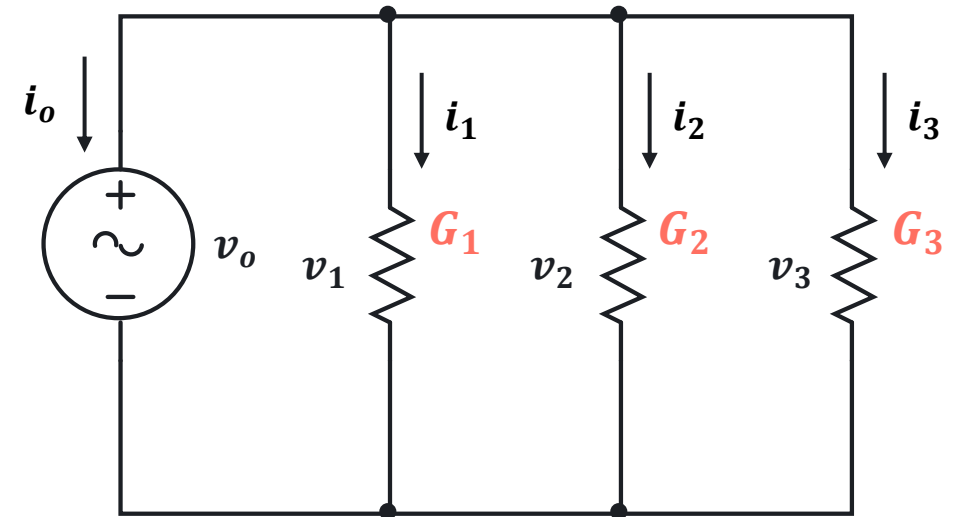
CONDUCTANCE

In a parallel network, the total conductance is the sum of the individual conductance of each resistor.

Mathematical Representation

$$G_o = G_1 + G_2 + G_3 + \cdots G_n$$

Parallel Network



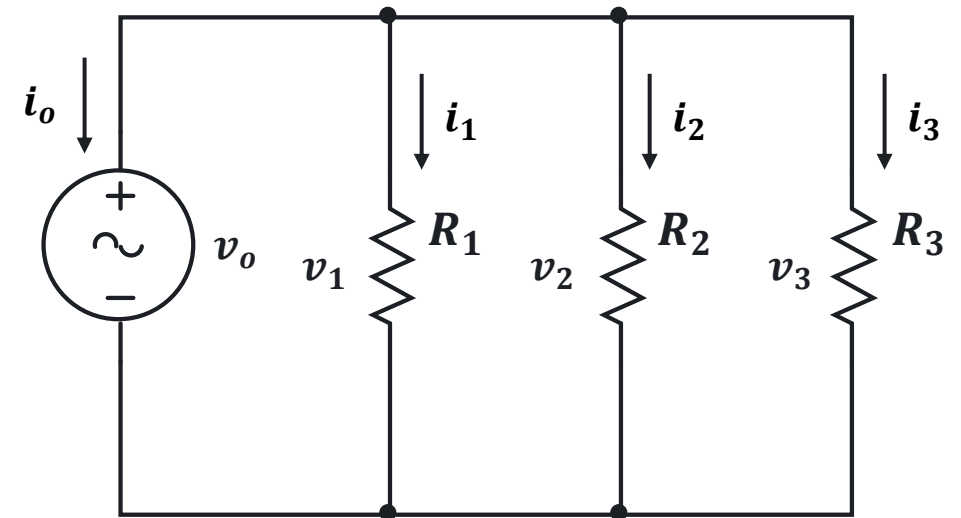
CURRENT

In a parallel network, the total current is the sum of the current flowing through each individual component.

Mathematical Representation

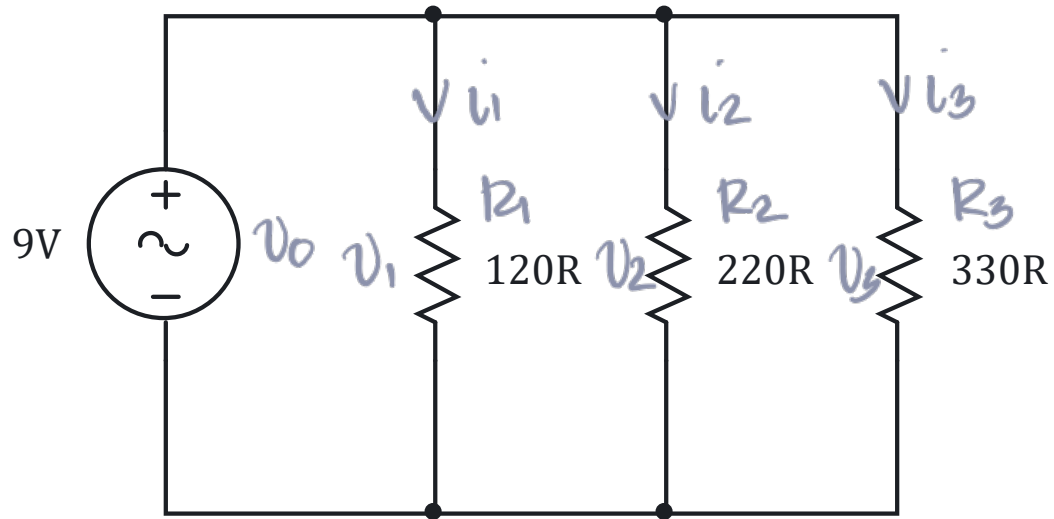
$$i_o = i_1 + i_2 + i_3 + \cdots i_n$$

Parallel Network



EXERCISE

Determine the current flowing through each resistor of the given circuit.



Solution

$$V_0 = V_1 = V_2 = V_3 = 9V$$

$$i_1 = \frac{V_1}{R_1}$$

$$i_1 = \frac{9}{120}$$

$$i_1 = 75 \text{ mA}$$

ans

$$i_2 = \frac{V_2}{R_2}$$

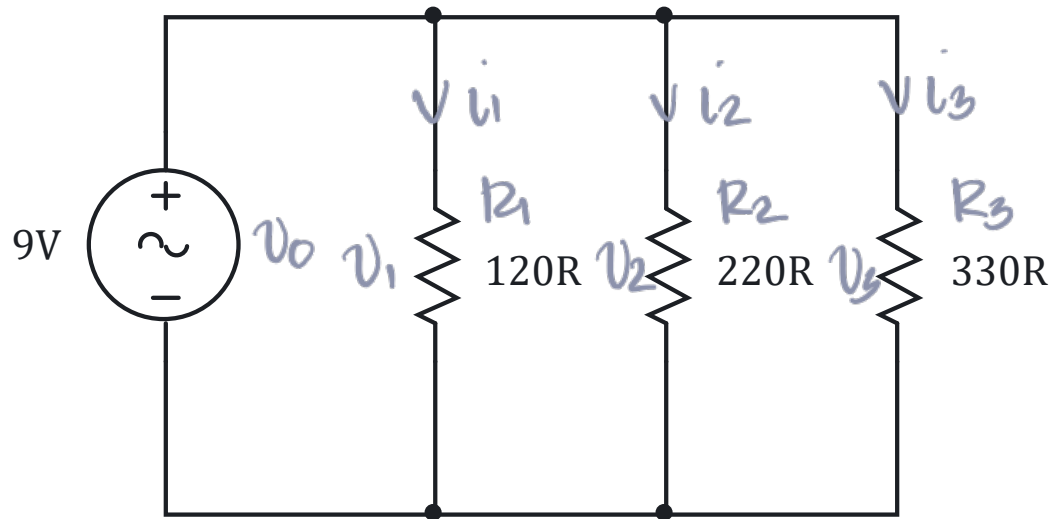
$$i_2 = \frac{9}{220}$$

$$i_2 = 40.91 \text{ mA}$$

ans

EXERCISE

Determine the current flowing through each resistor of the given circuit.



Solution

$$V_0 = V_1 = V_2 = V_3 = 9V$$

$$i_3 = \frac{V_3}{R_3}$$

$$i_3 = \frac{9}{330}$$

$$i_3 = 27.27 \text{ mA}$$

ans



SERIES PARALLEL NETWORK



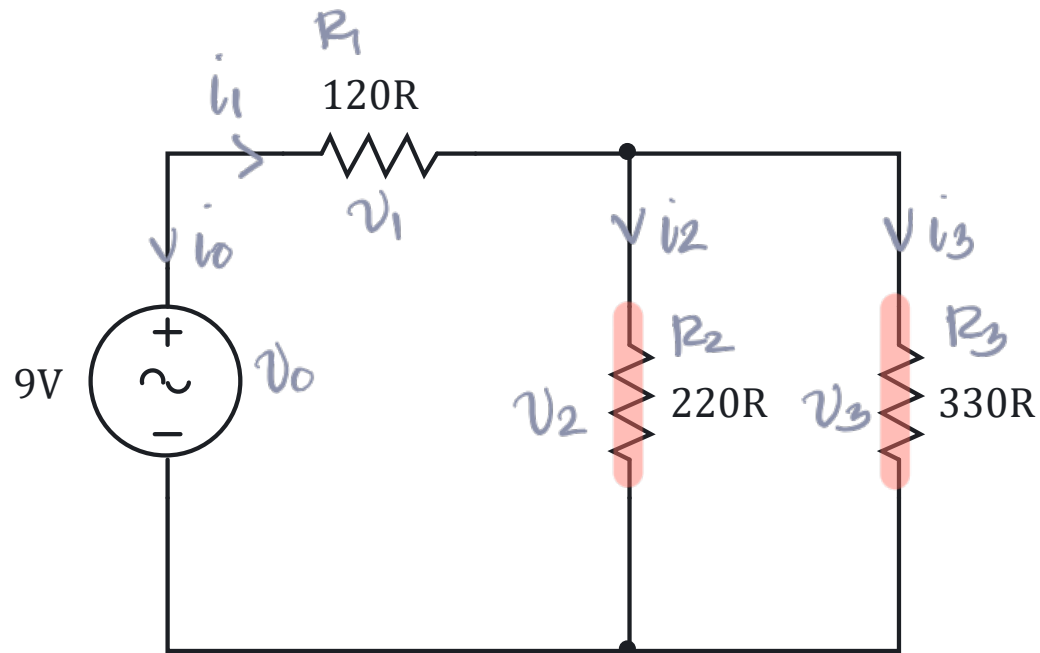
SERIEL-PARALLEL NETWORK

A series-parallel network is a type of electrical network that combines elements of both series and parallel circuits. These networks are commonly used in electrical and electronic systems to achieve desired voltage, current, and resistance characteristics.



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

Equivalent Resistance of R_2 and R_3

$$G_{2,3} = G_2 + G_3$$

$$\frac{1}{R_{2,3}} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{2,3}} = \frac{1}{220} + \frac{1}{330}$$

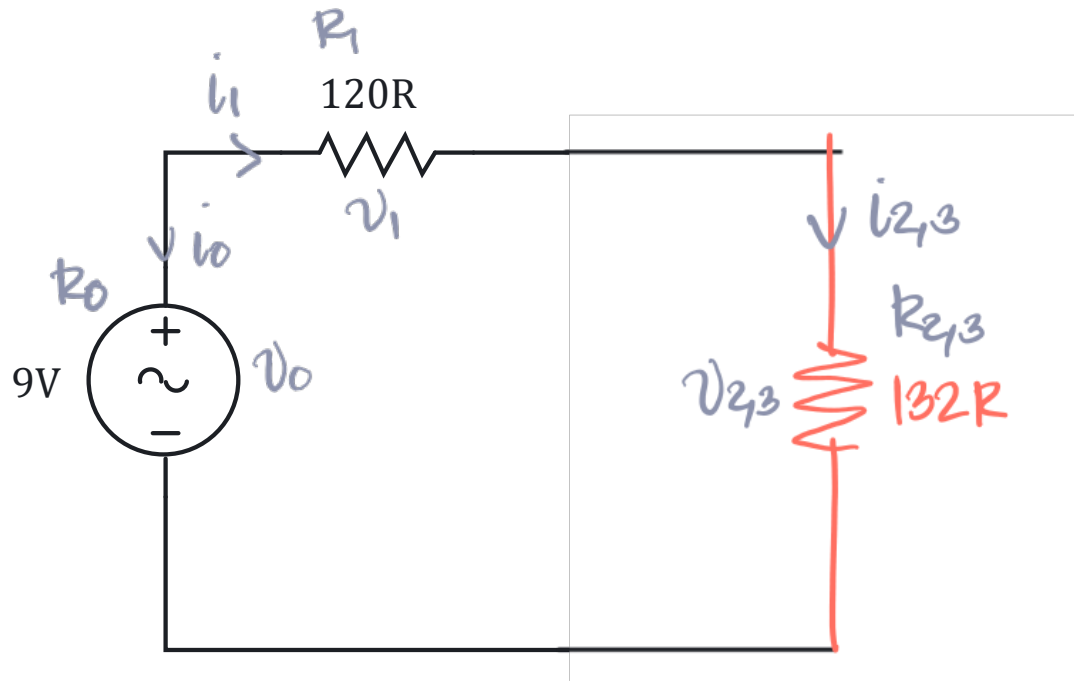
$$\frac{1}{R_{2,3}} = \frac{1}{132}$$

$$\underline{R_{2,3} = 132 \Omega}$$



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

Total Resistance

$$R_0 = R_1 + R_{2,3}$$

$$R_0 = 120 + 132$$

$$R_0 = 252 \Omega$$

Total Current

$$i_0 = \frac{v_0}{R_0}$$

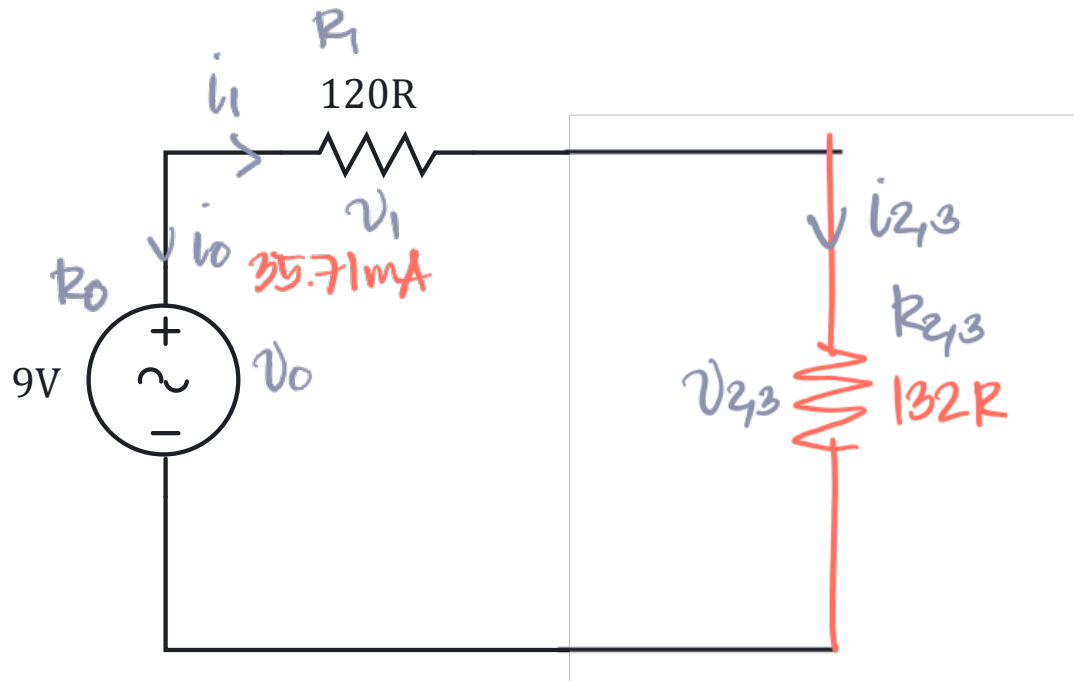
$$i_0 = \frac{9}{252}$$

$$i_0 = 35.71 \text{ mA}$$



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

$$i_0 = i_1 = i_{2,3}$$

$$i_1 = 35.71 \text{ mA}$$

ans

$$v_1 = i_1 R_1$$

$$v_1 = 35.71 \text{ m}(120)$$

$$v_1 = 4.29 \text{ V}$$

ans

$$v_{2,3} = i_{2,3} R_{2,3}$$

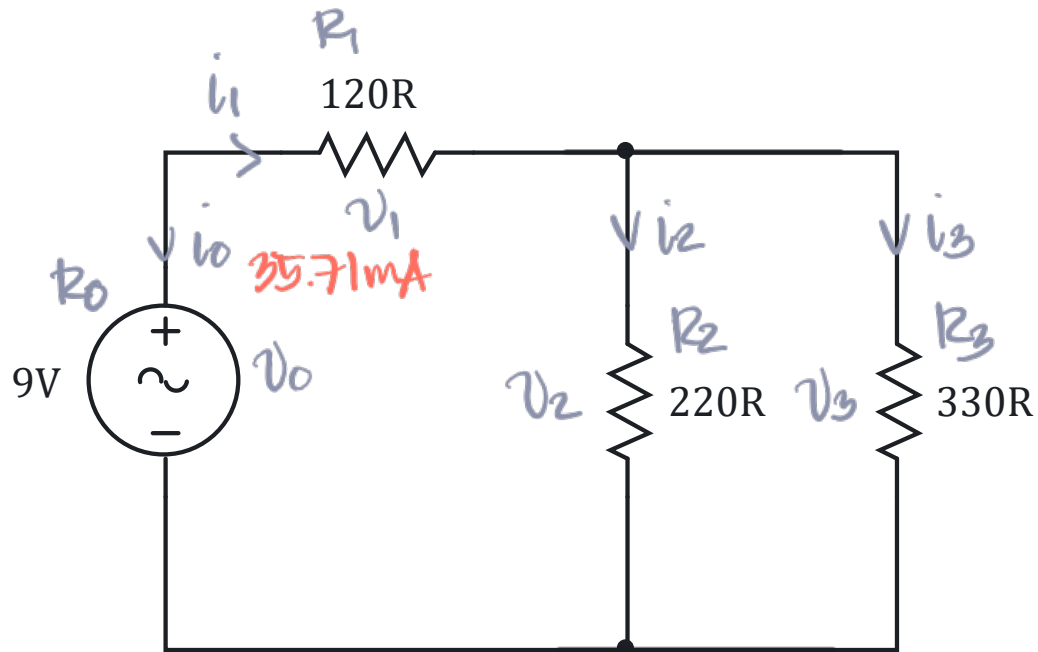
$$v_{2,3} = 35.71 \text{ m}(132)$$

$$v_{2,3} = 4.71 \text{ V}$$



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

$$v_{2,3} = v_2 = v_3$$

$$v_2 = 4.71 \text{ V}$$

ans

$$v_3 = 4.71 \text{ V}$$

ans

$$i_2 = \frac{v_2}{R_2}$$

$$i_2 = \frac{4.71}{220}$$

$$i_2 = 21.41 \text{ mA}$$

ans

$$i_3 = \frac{v_3}{R_3}$$

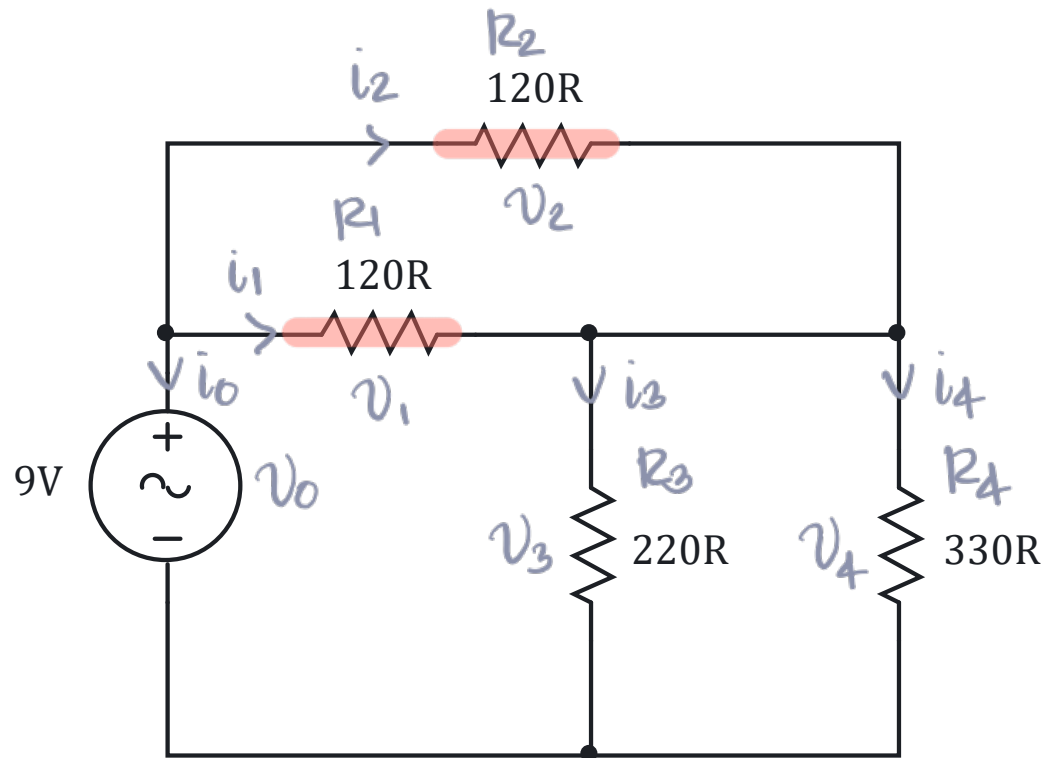
$$i_3 = \frac{4.71}{330}$$

$$i_3 = 14.71 \text{ mA}$$

ans

EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

Equivalent resistance of R_1 and R_2

$$G_{1,2} = G_1 + G_2$$

$$\frac{1}{R_{1,2}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{1,2}} = \frac{1}{120} + \frac{1}{120}$$

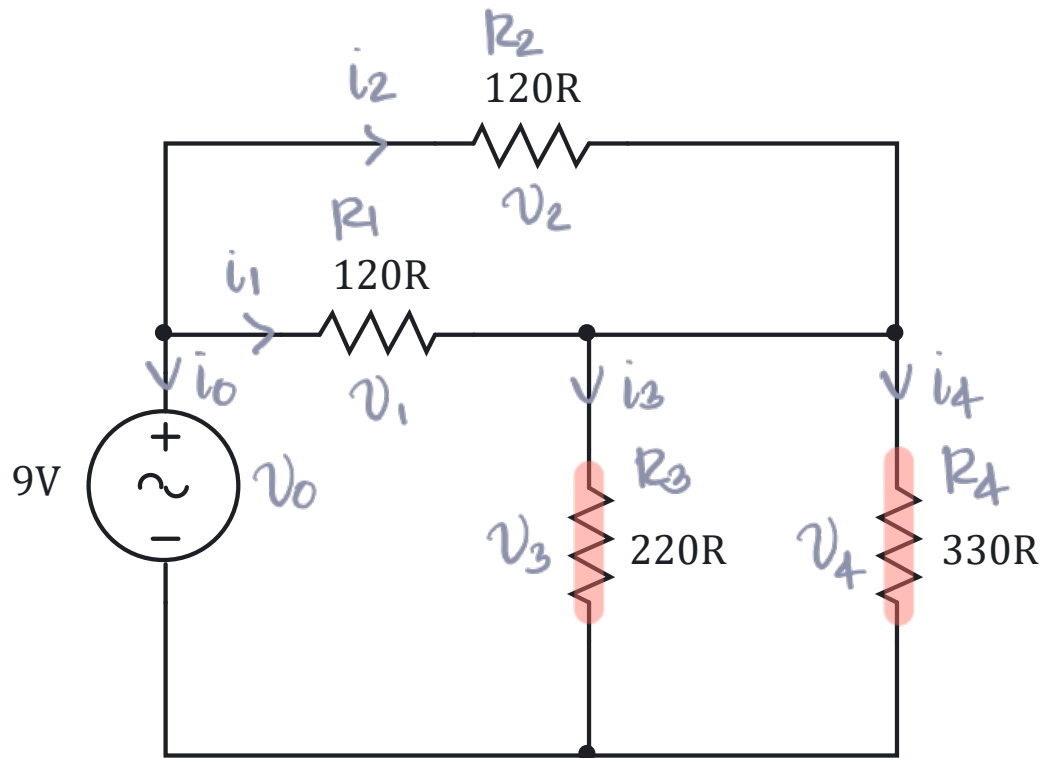
$$\frac{1}{R_{1,2}} = \frac{1}{60}$$

$$\underline{R_{1,2} = 60\Omega}$$



EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

Equivalent Resistance of R_3 and R_4

$$G_{3,4} = G_3 + G_4$$

$$\frac{1}{R_{3,4}} = \frac{1}{R_3} + \frac{1}{R_4}$$

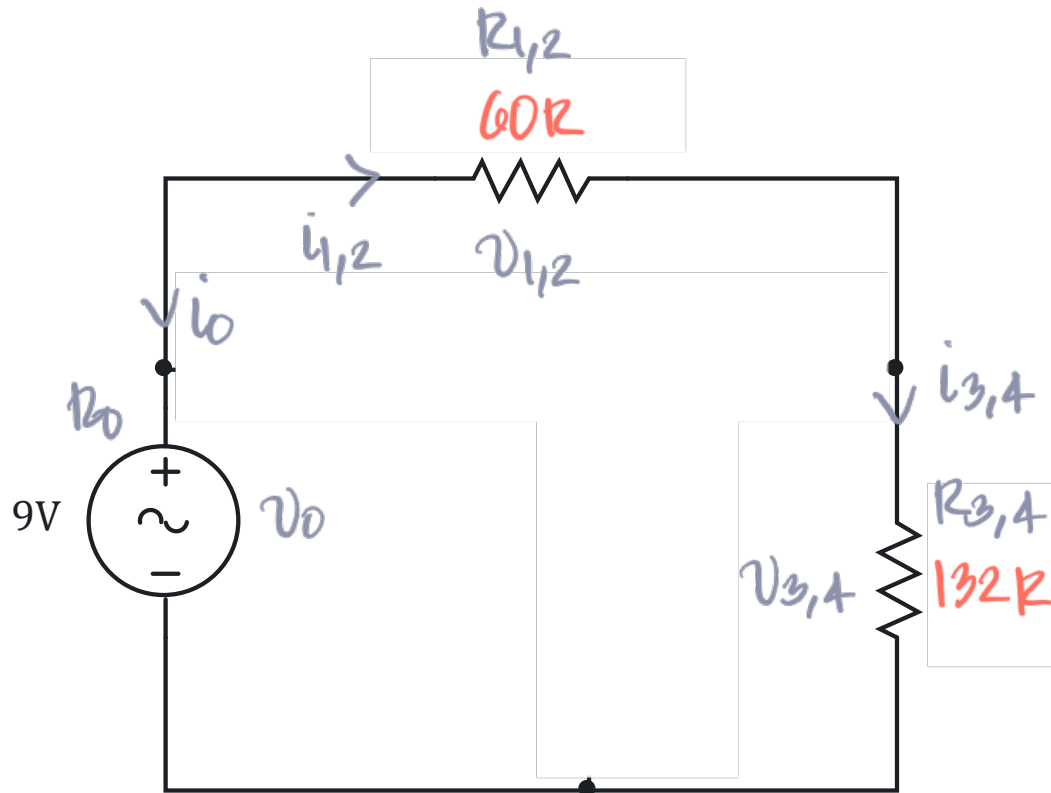
$$\frac{1}{R_{3,4}} = \frac{1}{220} + \frac{1}{330}$$

$$\frac{1}{R_{3,4}} = \frac{1}{132}$$

$$\underline{R_{3,4} = 132 \, \Omega}$$

EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

Total Resistance

$$R_0 = R_{1,2} + R_{3,4}$$

$$R_0 = 60 + 132$$

$$R_0 = 192$$

Total Current

$$i_0 = \frac{V_0}{R_0}$$

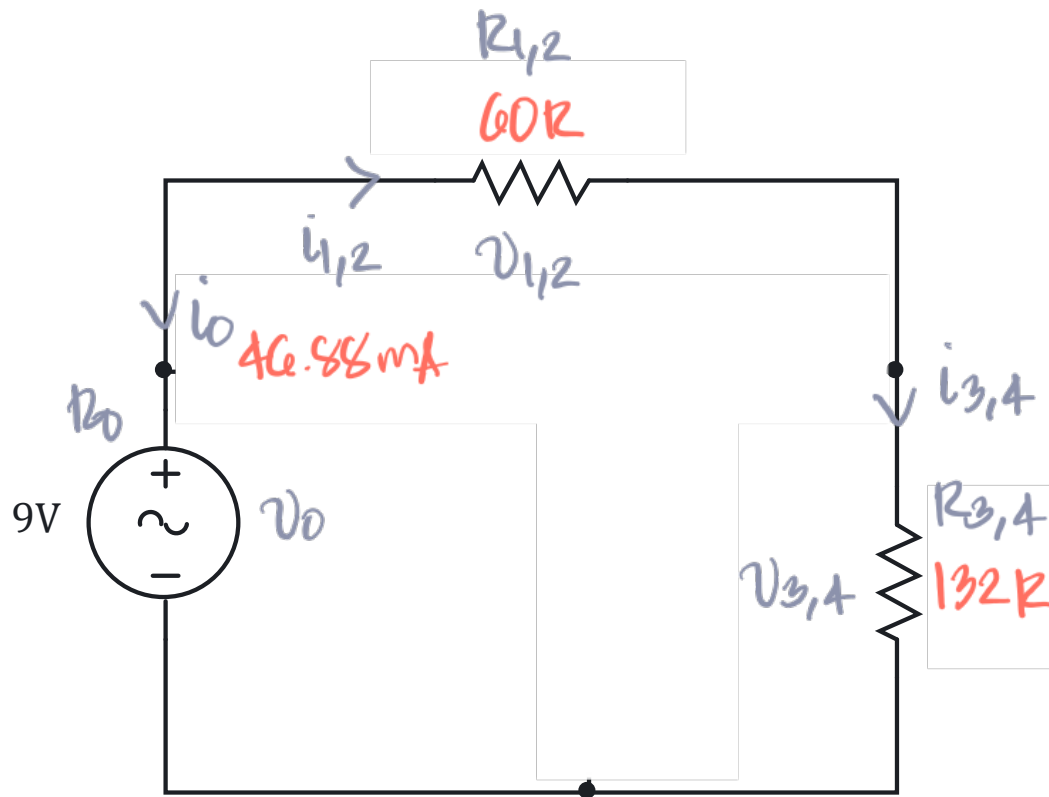
$$i_0 = \frac{9}{192}$$

$$i_0 = 46.88 \text{ mA}$$



EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

$$i_0 = i_{1,2} = i_{3,4}$$

$$V_{1,2} = i_{1,2} R_{1,2}$$

$$V_{1,2} = 46.88 \text{ mA} (60)$$

$$\underline{V_{1,2} = 2.81 \text{ V}}$$

$$V_{3,4} = i_{3,4} R_{3,4}$$

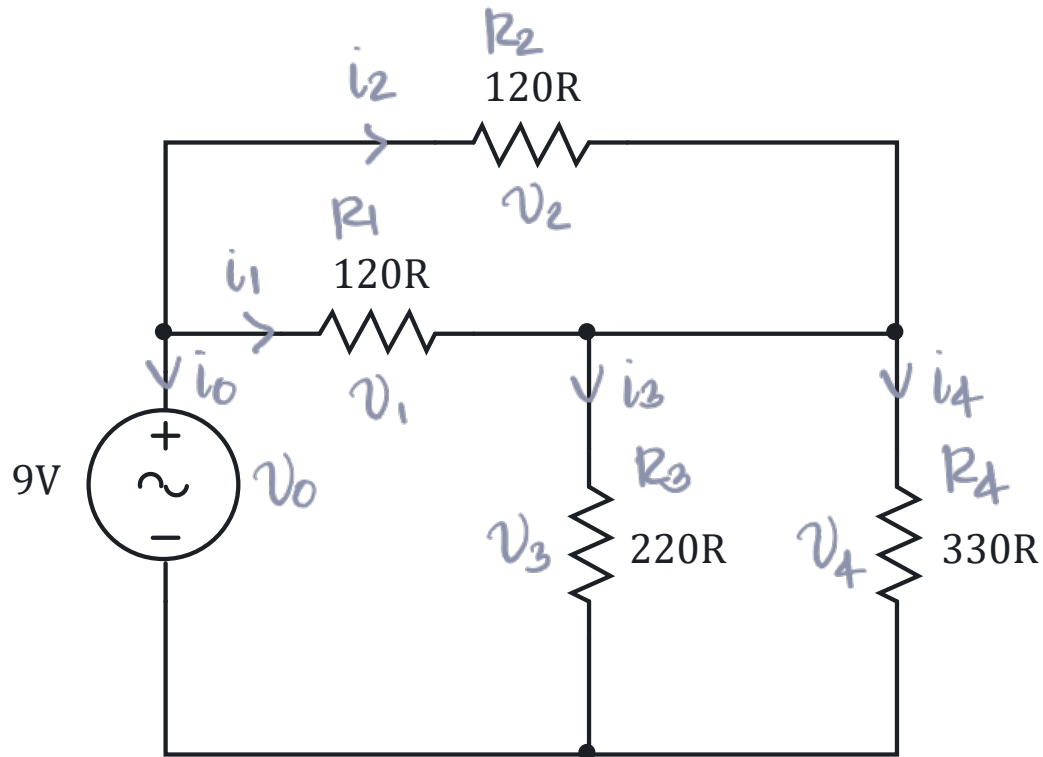
$$V_{3,4} = 46.88 \text{ mA} (132)$$

$$\underline{V_{3,4} = 6.19 \text{ V}}$$



EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

$$v_{1,2} = v_1 = v_2$$

$$v_1 = 2.81\text{ V}$$

ans

$$v_2 = 2.81\text{ V}$$

ans

$$i_1 = \frac{v_1}{R_1}$$

$$i_1 = \frac{2.81}{120}$$

$$i_1 = 23.42\text{ mA}$$

ans

$$i_2 = \frac{v_2}{R_2}$$

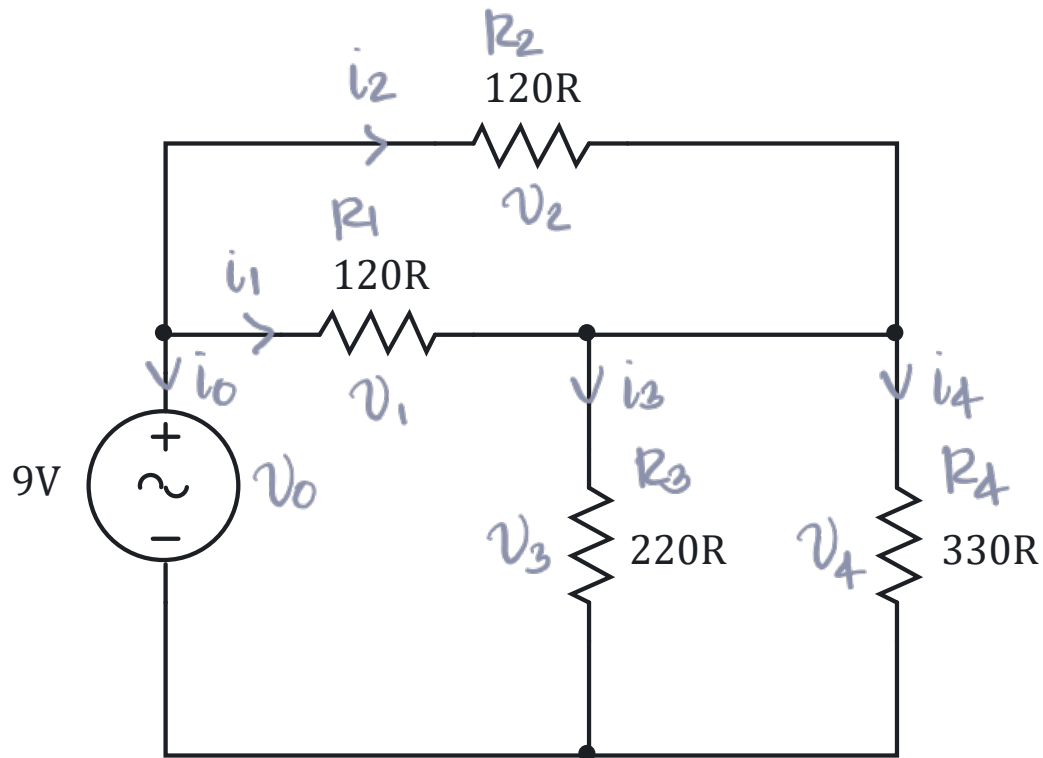
$$i_2 = \frac{2.81}{120}$$

$$i_2 = 23.42\text{ mA}$$

ans

EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

$$V_{3,4} = V_3 = V_4$$

$$V_3 = 6.19 \text{ V}$$

ans

$$V_4 = 6.19 \text{ V}$$

ans

$$i_3 = \frac{V_3}{R_3}$$

$$i_3 = \frac{6.19}{220}$$

$$i_3 = 28.14 \text{ mA}$$

ans

$$i_4 = \frac{V_4}{R_4}$$

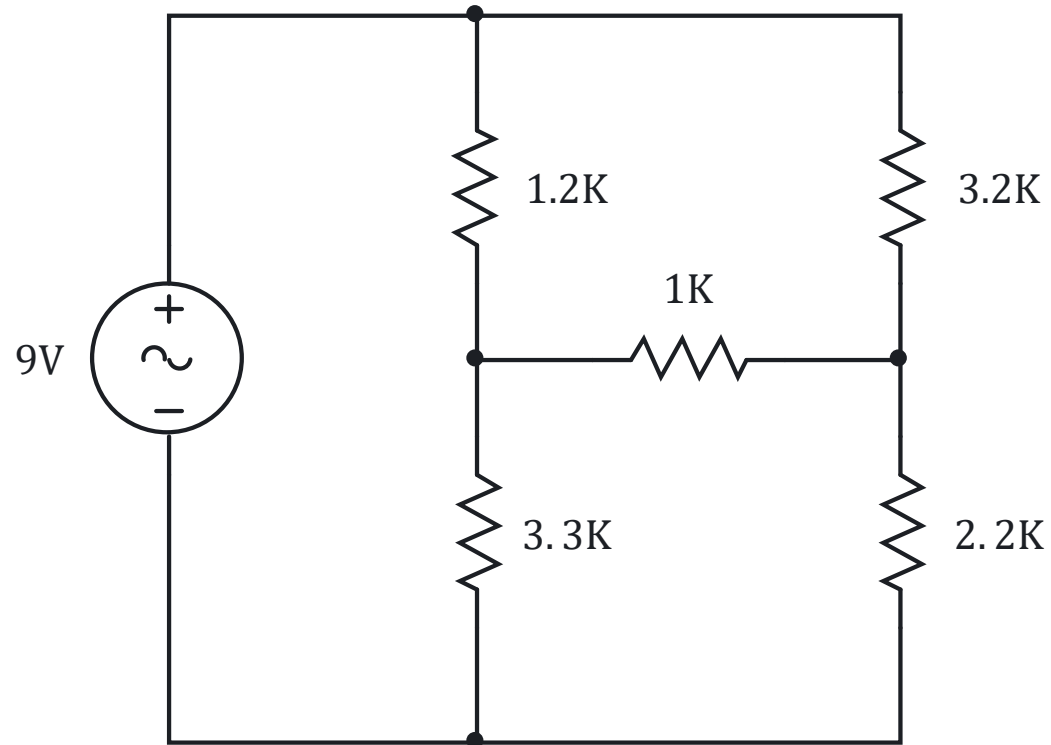
$$i_4 = \frac{6.19}{330}$$

$$i_4 = 18.76 \text{ mA}$$

ans

EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.



Solution

Solve using KCL/KVL



LABORATORY

